

What 320,000 Public Reviews Say About AI Conferences

Growth, Topic Shifts, Score Cutoffs, and Rebuttals at ICLR, NeurIPS, and ICML
(2018–2026)

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Abstract

We collected every public review, rebuttal, and decision available on OpenReview for 16 venue-years of the three largest machine-learning conferences: ICLR 2018–2026, NeurIPS 2021–2025, and ICML 2025–2026. The corpus has 84,212 papers and 322,903 reviews. Because ICLR publishes reviews of rejected and withdrawn papers as well, most of our analysis uses ICLR. We report six findings. (1) ICLR submissions grew 19-fold in eight years, from 1,018 to 19,814, and in 2026 almost one submission in three (31%) was withdrawn or desk-rejected rather than decided. (2) Papers with “LLM” or “language model” in the title rose from 1.0% to 21.4% of submissions, with a four-fold jump in the first cycle after ChatGPT. GAN titles nearly vanished, diffusion took their place, and “reasoning” and “agent” grew 4.6- and 3.1-fold between 2024 and 2026. Yet LLM papers were accepted at exactly the overall rate (39%). (3) The weakness raised most often is missing or weak baselines (33.5% of reviews), followed by scalability and compute cost (31.1%). Novelty comes fifth (22.1%). (4) Acceptance is a steep function of the mean score. At ICLR 2026, a mean of 5.0 out of 10 was accepted 54% of the time, and each half point between 4.5 and 5.5 moved the probability by about 25 percentage points. (5) Only 5 of 1,209 decided papers without any author reply were accepted, compared with 38% of papers with a reply and 61% of papers where a reviewer wrote that they raised their score. This association is not causal. (6) From 2024 to 2026 the number of ICLR reviews grew 2.7-fold while the median review length fell from 278 to 249 words. The share of reviews containing “delve” rose nine-fold from 2022 to 2024 and then fell back to the 2022 level.

1 Introduction

Peer review at machine-learning conferences has become a large-scale system. Thousands of volunteer reviewers score tens of thousands of submissions in a few weeks, and many authors experience the outcome as noisy [1, 4]. OpenReview [5] makes an unusual amount of this process public: the reviews, scores, author responses, reviewer follow-ups, and final decisions. Earlier datasets such as PeerRead [2] covered a few thousand papers. The public record now covers hundreds of thousands of reviews, enough to measure how the system behaves rather than how it is supposed to behave.

This note is descriptive. We do not propose a model of reviewing. We count, as transparently as we can, what the public record shows about four practical questions an author might ask. What are people submitting? What do reviewers object to? What mean score is enough? Does the

rebuttal matter? We also report two signals about the reviews themselves: their length, and the frequency of words that are characteristic of LLM-generated text [3].

2 Data

We used the OpenReview API (v2, falling back to v1 for older venues) to download, for every submission, its title, abstract, keywords, and all replies: official reviews, meta-reviews, decisions, author comments or rebuttals, reviewer comments, and withdrawal notices. PDFs were not downloaded. Collection finished on 7 October 2026.

Venue	Papers	Reviews	What is public
ICLR 2018–2026 (9 years)	55,472	209,348	all submissions, including rejected and withdrawn
NeurIPS 2021–2025 (5 years)	18,763	75,254	accepted + rejected papers whose authors opted in
ICML 2025–2026 (2 years)	9,977	38,301	accepted + rejected papers whose authors opted in
Total (16 venue-years)	84,212	322,903	

Table 1: The corpus. Reviews are official reviews with non-empty text. NeurIPS Datasets & Benchmarks tracks and ICML 2023–2024 (reviews not public) are excluded from the totals.

At NeurIPS and ICML, roughly 95% of the papers with public reviews are accepted. This reflects the publication policy, not the true acceptance rate. Any statistic that compares accepted and rejected papers is therefore computed on ICLR, where the full population is public.

Decisions. A paper is *Withdrawn* or *Desk rejected* if it has the corresponding note or venue string. Otherwise its decision comes from the decision note, or from the meta-review recommendation in ICLR 2019–2022. “Invite to workshop” in 2018 counts as rejected. In ICLR 2018, 95 submissions have no decision and no withdrawal note; these are withdrawn v1 submissions and are shown as withdrawn in Figure 1.

Scores. Scores are the latest version stored on OpenReview, so they include any change a reviewer made after the rebuttal. Scales differ across venues and years. ICLR 2026 uses $\{0, 2, 4, 6, 8, 10\}$, while ICLR 2024–2025 use $\{1, 3, 5, 6, 8, 10\}$. We therefore report score cutoffs for one venue-year only.

Keyword measures. Topic, weakness, “score raised”, and LLM-style word counts are regular-expression matches. These measures are approximate. We read samples of matches to remove obvious false positives: for example, “information theory” was initially counted as a theory complaint. Appendix A lists the exact patterns.

3 Findings

3.1 Growth, and a third of submissions that are never decided

ICLR received 1,018 submissions in 2018 and 19,814 in 2026, a 19-fold increase (Figure 1). The last two cycles grew by $1.6\times$ and $1.7\times$. The share of submissions accepted has stayed in a narrow 26–33% band. What changed is the share that never reaches a decision. In 2026, 26.3% of submissions were withdrawn and 4.6% were desk-rejected. Desk rejections rose from 53 in 2024 and 71 in 2025 to 908 in 2026.

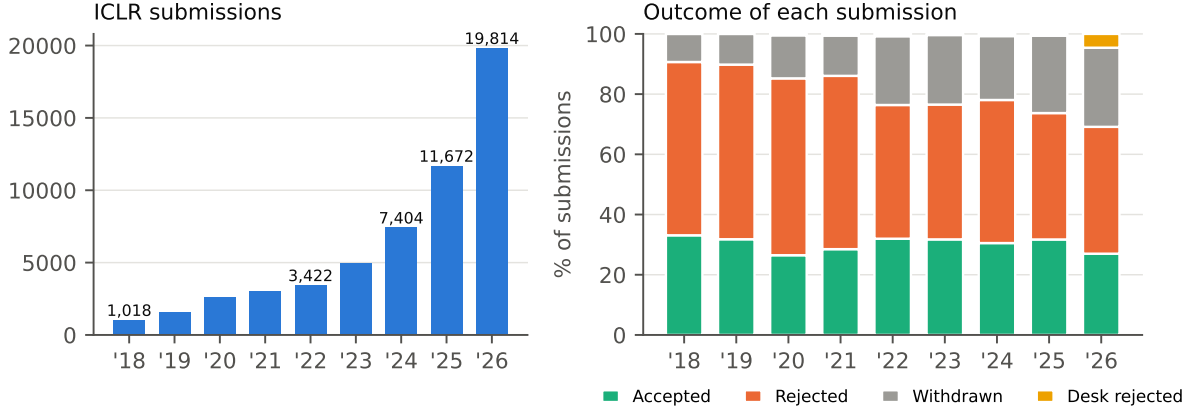


Figure 1: ICLR submissions per year (left) and the outcome of each submission (right). ICLR 2018 submissions without any decision or withdrawal note (9%) are counted as withdrawn.

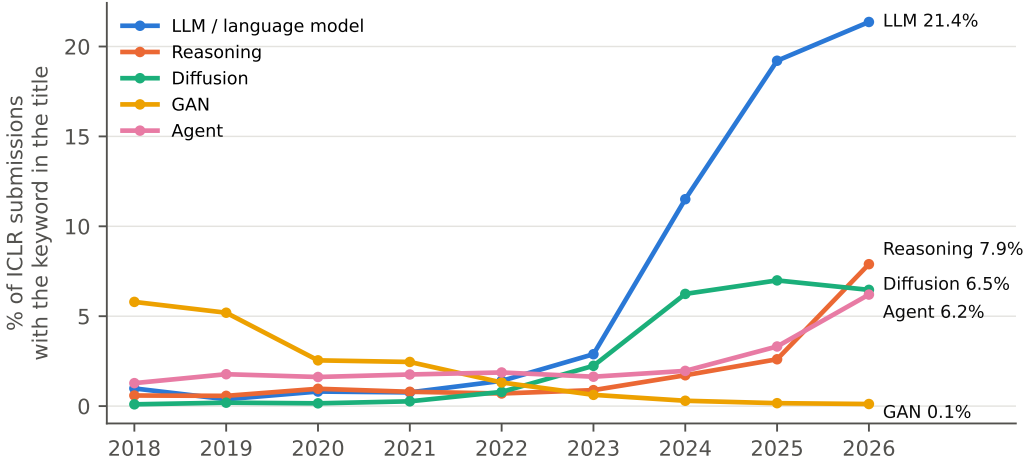


Figure 2: Share of ICLR submissions (including withdrawn and desk-rejected) whose title matches each keyword.

3.2 Topics: LLMs everywhere, and GANs nowhere

Figure 2 tracks the share of ICLR submissions whose *title* contains a keyword. Titles mentioning LLMs or language models stayed between 0.4% and 1.4% until 2022. They then rose to 2.9% in 2023, 11.5% in 2024, and 21.4% in 2026, so roughly one submission in five. The jump from 2023 to 2024 is four-fold. The ICLR 2024 deadline (September 2023) was the first after ChatGPT’s public release in November 2022.

Generative modelling changed over completely. GAN titles fell from 5.8% (2018) to 0.1% (2026), while diffusion and flow-matching titles rose from 0.8% (2022) to 7.0% (2025). The fastest-growing keywords now are *reasoning* (1.7% $\rightarrow$ 7.9%, 4.6 $\times$ from 2024 to 2026) and *agent* (2.0% $\rightarrow$ 6.2%, 3.1 $\times$).

Popularity does not translate into acceptance. Among decided ICLR 2026 papers, the acceptance rate was 39.2% for LLM titles, against 39.0% overall. Diffusion titles were accepted at 45.8%, reasoning at 44.0%, and agent at 36.6%.

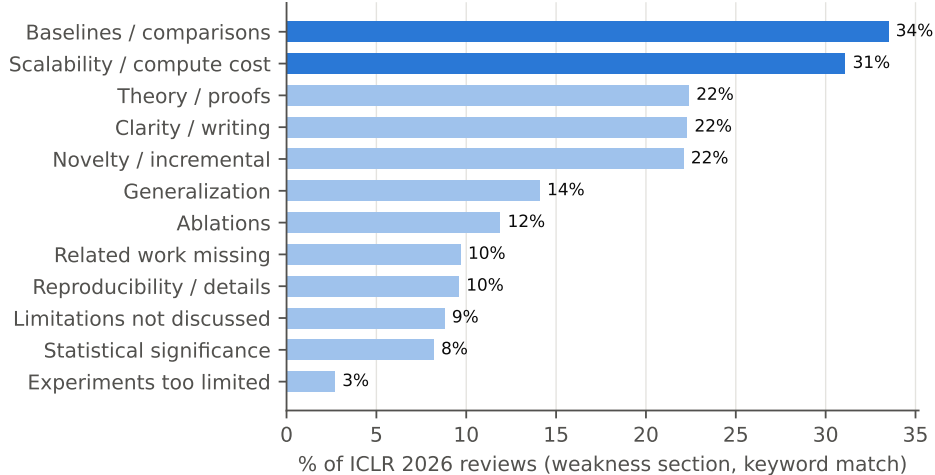


Figure 3: Share of ICLR 2026 reviews ($n = 75,859$) whose weaknesses section matches each theme. A review can match several themes.

3.3 What reviewers object to

For each ICLR 2026 review we matched twelve themes against the weaknesses section (Figure 3). The most common complaint concerns baselines and comparisons (33.5% of reviews), followed by scalability and computational cost (31.1%). Theory, clarity, and novelty form a close second tier at 22.4%, 22.3%, and 22.1%. In other words, reviewers ask “is it better than what exists, and at what cost?” about one and a half times as often as they ask “is it new?”.

3.4 Score cutoffs

The mean score is 5.40 for accepted and 3.97 for rejected ICLR 2026 papers. Figure 4 (left) shows the acceptance rate of the 13,697 decided papers, grouped by mean score rounded to the nearest 0.5. The curve is steep: 12% at 4.0, 29% at 4.5, 54% at 5.0, 79% at 5.5, and 93% at 6.0. With four reviewers on the $\{0, 2, 4, 6, 8, 10\}$ scale, one reviewer moving from 4 to 6 raises the mean by exactly 0.5. Near the middle of the curve, that is a change of about 25 percentage points in the chance of acceptance.

3.5 Rebuttals

At ICLR 2026, the authors of 73.8% of papers posted a reply, and the median reply (476 words) was about twice as long as the median review (248 words). Among decided papers (Figure 4, right), 5 of the 1,209 papers without any author reply were accepted (0.4%). Of the 9,865 papers with a reply but no reviewer statement of a raised score, 37.9% were accepted. Of the 2,623 papers where a reviewer wrote that they raised their score (19% of decided papers), 61.2% were accepted.

These numbers describe an association, not the effect of writing a rebuttal. Authors who expect rejection often do not reply, and scores in our data are final scores. Still, the arithmetic of the previous subsection explains why the rebuttal phase matters so much near the threshold: convincing a single reviewer is enough to move a borderline paper across it.

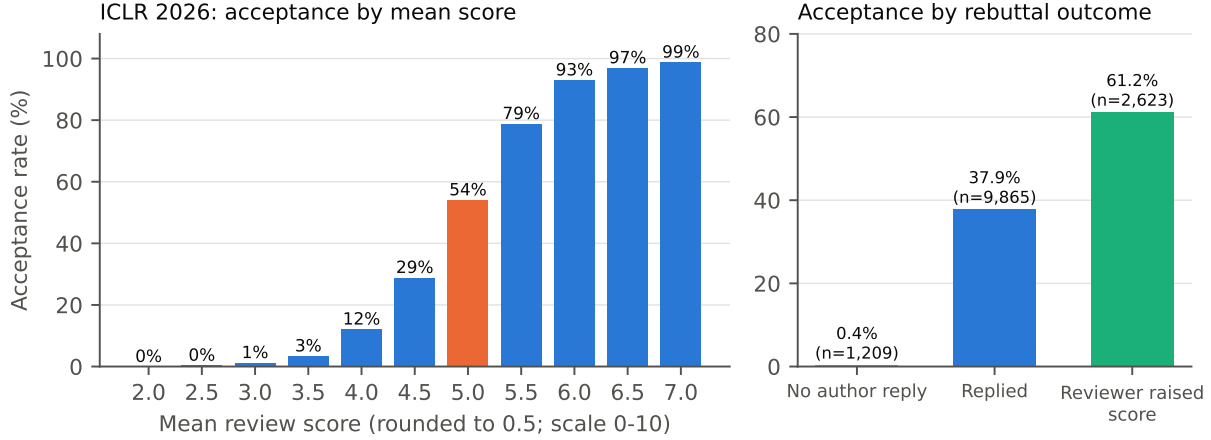


Figure 4: Left: ICLR 2026 acceptance rate by mean review score (final scores, rounded to 0.5). Right: acceptance rate by rebuttal outcome. “Reviewer raised score” means at least one reviewer comment states that the reviewer raised, or will raise, their score; conditional sentences are excluded.

3.6 The reviews themselves

Between 2024 and 2026, ICLR used the same review form (summary, strengths, weaknesses, questions), so lengths are comparable. Over these years the number of reviews grew from 28,002 to 75,781, while the median review length fell from 278 to 249 words (Figure 5, left). We do not compare with earlier years, whose forms differ.

Liang et al. [3] showed that LLM-modified text shows up as a sharp rise in certain adjectives and verbs. We tracked four of them (Figure 5, right). The share of ICLR reviews containing “delve” was 0.04–0.18% through 2023, rose to 1.56% in 2024 (nine times the 2022 level), and fell to 0.33% in 2025 and 0.10% in 2026. “Meticulous” rose eleven-fold over the same window. NeurIPS reviews moved the same way: “delve” went from 0.18% in 2022 to 0.54% in 2023. The decline after 2024 does not by itself show that reviewers used LLMs less. Newer models write differently, and word-frequency signals are easy to evade.

4 Limitations

All topic, weakness, and word measures are keyword matches. They miss paraphrases and include some false positives. Titles also capture only how authors chose to frame a paper. The rebuttal analysis is observational and uses final scores. “Score raised” is detected from free text and undercounts silent score changes. NeurIPS and ICML expose only a self-selected subset of rejected papers, so we do not compare acceptance across venues. Finally, OpenReview content can be edited or removed, so the numbers describe the record as of 7 October 2026.

5 Conclusion

Three practical observations follow for authors. First, reviewers ask about baselines and cost more often than about novelty, so comparisons and compute budgets deserve as much care as the idea. Second, near the acceptance threshold one reviewer’s score is worth about 25 percentage points, which makes a careful rebuttal one of the highest-leverage steps in the process. Third, following

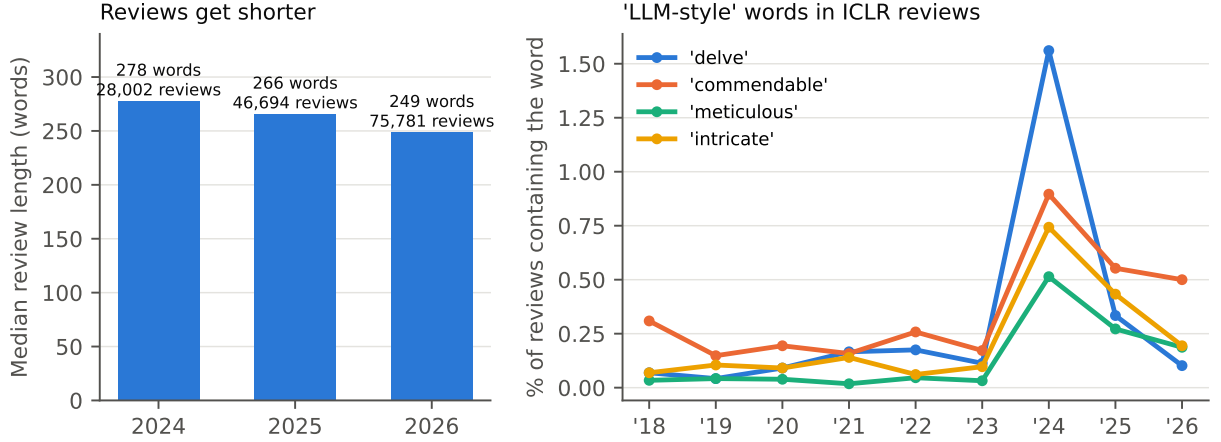


Figure 5: Left: number and median length of ICLR reviews, 2024–2026 (same review form). Right: share of ICLR reviews containing four words associated with LLM-generated text [3].

the trend does not raise the odds. LLM papers are accepted at the average rate and face the most competition.

Statement on AI assistance. The data-collection and analysis code, the figures, and drafts of this text were prepared with the help of an AI assistant (Claude, Anthropic). All numbers were computed directly from the OpenReview data and checked programmatically against the aggregated statistics. The author chose the questions and is responsible for the interpretation.

References

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A Keyword patterns

All patterns are case-insensitive Python regular expressions.

- **LLM:** `\b(llms?|mlms?|vlms?|large language models?|language models?|gpt)\b`
- **Diffusion:** `diffusion|flow matching`; **GAN:** `\bgans?\b|generative adversarial`

- **Reasoning:** reasoning; **Agent:** \bagents?\b|agentic
- **Baselines:** baseline|compar(e|ison) (with|to|against)|state[-]of[-]the[-]art|sota
- **Scalability:** scal(e|ab)|computational (cost|overhead|complexity)|runtime|efficien
- **Novelty:** novel|incremental|limited contribution|not new
- **Score raised** (reviewer comments, per sentence, excluding sentences that contain *if*, *would*, *could*, *might*, *consider*, *willing*, *open to*):
\ b(rais|increas|bump|upgrad)\w*\s+(up\s+)?(my|the)\s+(overall\s+)?(score|rating)s?\b

The remaining weakness themes and the full code are available from the author on request.